

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 307453359

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.740101 d
Epoch (t0) 1493.4111 BTJD
Duration 0.0840 d
Depth 7.111 % (71,106 ppm)
SNR 107.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

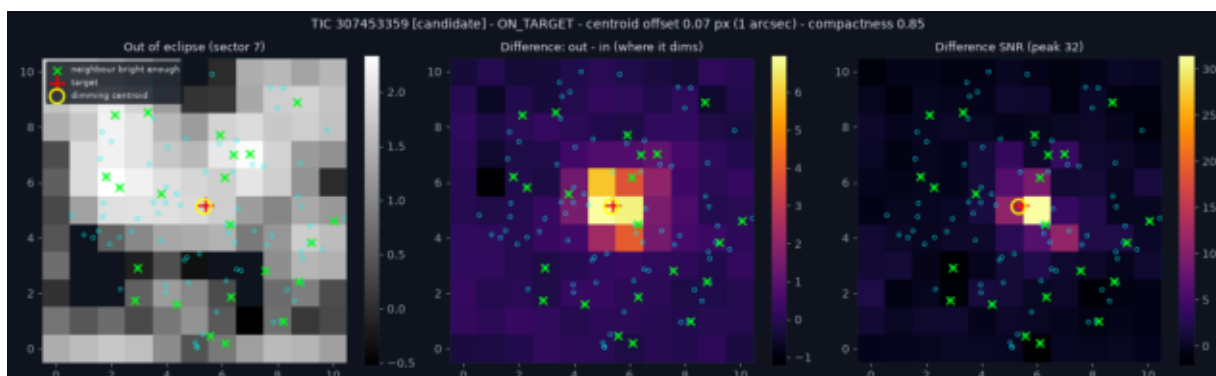
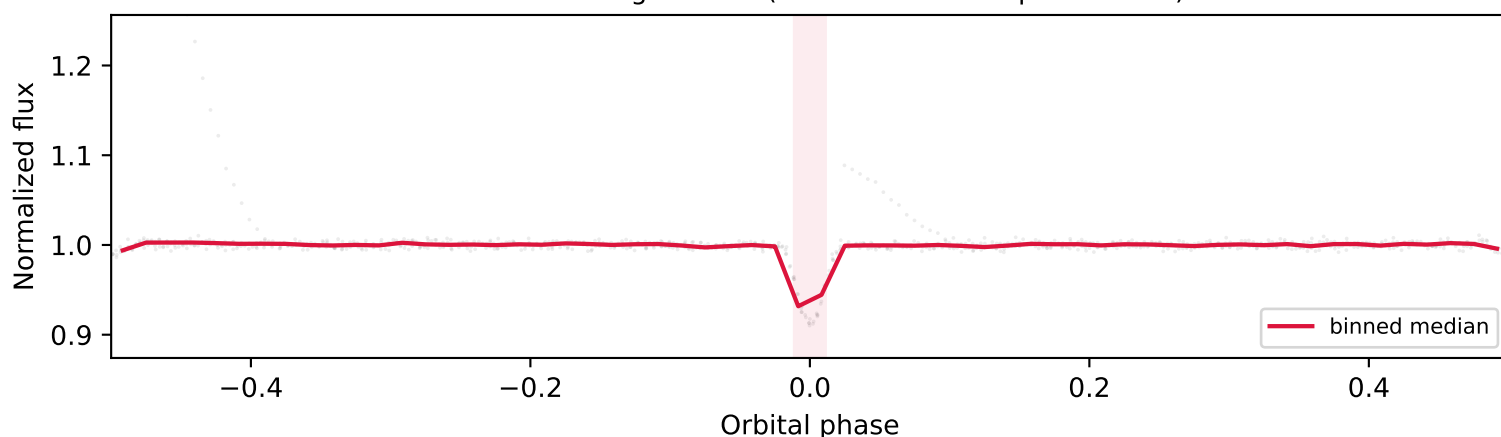
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 89.0 (gate ≥ 7.0)
Odd/even z-score 0.65 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 82.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 31.51 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 307453353
1.14 px away, dTmag +2.32 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 232261044

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	0.810821 d (ALIASED, see below)	VSX	no match
-> corrected period:	1.621641 d	ASAS-SN	no match
Epoch (t0)	1468.3491 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1200 d	Gaia variable classification	no match
Depth	2.534 % (25,343 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	88.5	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	23 (gate >= 3)		
Robust depth ratio	1.07 (gate >= 0.50)		
Robust depth sigma	53.0 (gate >= 7.0)		
Odd/even z-score	5.53 (ALIAS FLAGGED)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

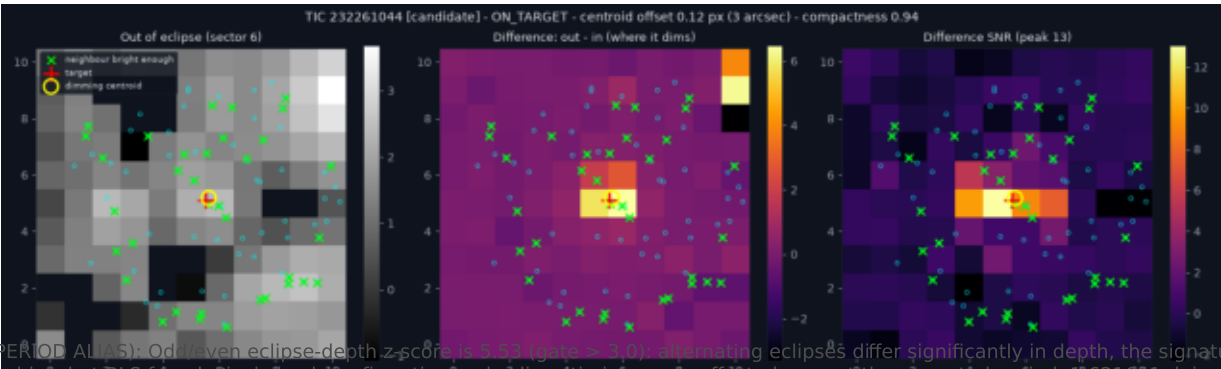
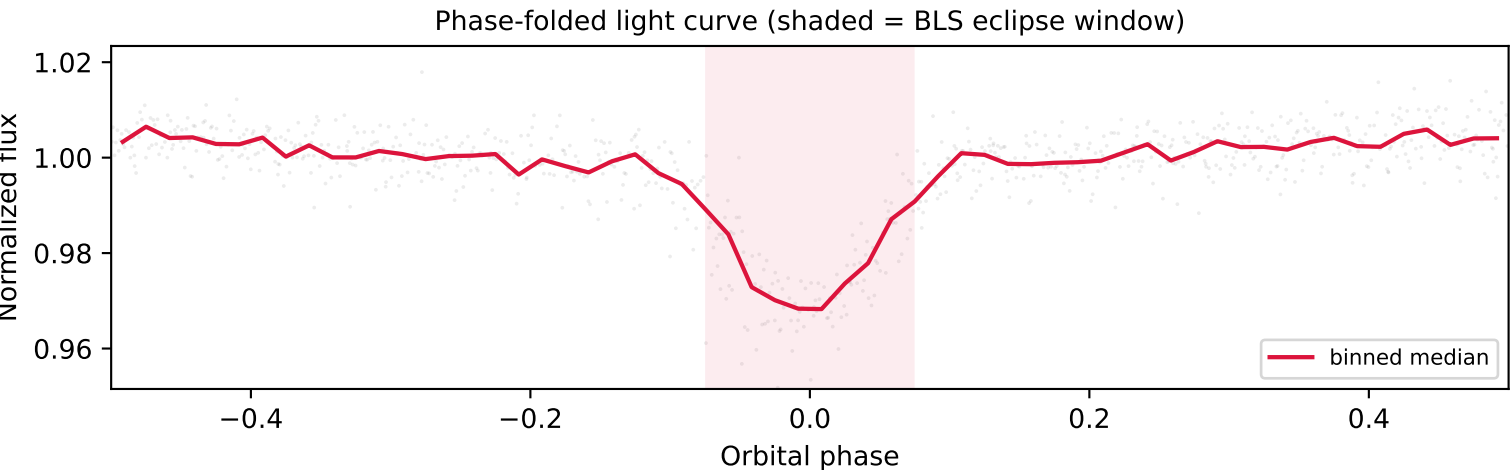
Sector 6 light-curve dip in this sector: 53.8 sigma t0 refined by 0.0025 phase

Difference-image peak SNR 13.06 (gate >= 5.0) Compactness 0.94 (gate >= 0.50)

Dimming centroid offset 0.12 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 713665868

0.20 px away, dTmag +0.56 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 5.53 (gate >= 7.0). Aliased eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel check confirmation and eclipse timing are unaffected - report the corrected period, 1.621641 d, in any write-up. Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.075 px (dimming centroid 0.125 px from target vs 0.200 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.